Lesson 5: Vector Spaces and Linear Maps

AI/ML Mathematics

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1 Motivation and Overview

Consider a neural network with a single hidden layer. The forward pass computes $x \mapsto \sigma(W_2\,\sigma(W_1 x + b_1) + b_2)$, where $W_1 \in \mathbb{R}^{d\times n}$ and $W_2 \in \mathbb{R}^{k\times d}$ are weight matrices. Strip away the nonlinearity σ for a moment: the remaining map $x \mapsto W_2 W_1 x$ is a *composition of linear maps* $\mathbb{R}^n \to \mathbb{R}^d \to \mathbb{R}^k$. Its range, null space, and rank determine which inputs the network can distinguish and how many independent features the hidden layer can encode. If $\operatorname{rank}(W_2 W_1) < \min(k, n)$, the network suffers a *bottleneck* – it collapses distinct inputs to the same output.

More broadly, policy gradient methods in reinforcement learning parameterize a policy π_θ by a vector $\theta \in \mathbb{R}^p$. The parameter space is a vector space, and the Fisher information matrix defines a natural inner product on its tangent space. Understanding bases, dimension, and change of basis is prerequisite to reasoning about such geometric structures. The tools developed in this lesson will let us define linear maps rigorously, prove the rank–nullity theorem, and analyze how matrix representations change with a choice of basis – all of which are essential for the spectral and optimization theory developed in later lessons.

Where Vector Spaces and Linear Maps Appear in RL/ML

- **Linear function approximation (TD learning).** Value functions are approximated as $\hat{V}(s) = \phi(s)^\top w$, a linear map from feature space to $\mathbb{R}$; the rank of the feature matrix Φ determines expressiveness.
- **Policy gradient / natural gradient.** The parameter update $\theta \leftarrow \theta + \alpha\, F^{-1}\nabla J$ requires understanding coordinate changes and similarity transformations.
- **Principal component analysis (PCA).** Dimensionality reduction projects data onto a subspace spanned by the top eigenvectors – a direct application of bases, span, and projections.

Prerequisites. Familiarity with sets, functions, and the field axioms of $\mathbb{R}$ (Lesson 1); sequences and limits in $\mathbb{R}$ (Lesson 2); continuity of real-valued functions (Lesson 3); basic properties of $\mathbb{R}^n$ as a set of tuples.

Learning objectives. After completing this lesson, the student will be able to:

- **Define** a vector space, subspace, span, linear independence, basis, dimension, and linear map, and verify these definitions in concrete examples.
- **Prove** that the intersection of subspaces is a subspace and that any two bases of a finite-dimensional space have the same cardinality.
- **Derive** the rank–nullity theorem from the definition of a basis and apply it to determine injectivity and surjectivity of linear maps.
- **Compute** the matrix representation of a linear map with respect to given bases and carry out a change-of-basis computation.
- **Classify** linear maps between finite-dimensional spaces as injective, surjective, or bijective using dimension arguments.

2 References

- Axler [2015, Chapters 1–3] – Primary reference; develops vector spaces, span, independence, bases, and linear maps without determinants.
- Hoffman and Kunze [1971, Chapters 1–3] – Classical treatment with emphasis on the interplay between abstract vector spaces and matrices.

- Strang [2016, Chapters 1–3] – Computational perspective with many worked examples connecting to applications.
- Horn and Johnson [2013, Chapter 1] – Concise review of vector spaces and linear maps as background for matrix analysis.
- Lax [2007, Chapters 1–4] – Elegant proofs and connections to functional analysis.

3 Vector Spaces

Throughout this lesson, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

3.1 Axioms

Definition 3.1 (Vector Space)**.** A *vector space* over a field $\mathbb{F}$ is a set V together with an addition $+\colon V \times V \to V$ and a scalar multiplication $\cdot\colon \mathbb{F} \times V \to V$ satisfying the following axioms for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$:

1. **Commutativity:** $u + v = v + u$.
2. **Associativity of addition:** $(u + v) + w = u + (v + w)$.
3. **Additive identity:** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
4. **Additive inverse:** For each $v \in V$, there exists $w \in V$ with $v + w = 0$.
5. **Multiplicative identity:** $1v = v$.
6. **Associativity of scalar multiplication:** $a(bv) = (ab)v$.
7. **Distributive laws:** $a(u + v) = au + av$ and $(a + b)v = av + bv$.

Elements of V are called *vectors* and elements of $\mathbb{F}$ are called *scalars*.

Remark 3.2. The additive identity is unique: if $0'$ also satisfies $v + 0' = v$ for all v, then $0 = 0 + 0' = 0'$. Similarly, each vector has a unique additive inverse, which we denote $-v$.

Example 3.3 (Standard Vector Spaces)**.** The following are vector spaces over $\mathbb{F}$:

1. $\mathbb{F}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{F}\}$ with componentwise addition and scalar multiplication.
2. $\mathbb{F}[x]_{\leq n}$, the set of polynomials with coefficients in $\mathbb{F}$ of degree at most n, with usual polynomial operations.
3. $C[a, b]$, the set of continuous functions $f\colon [a, b] \to \mathbb{R}$, with pointwise addition $(f + g)(x) = f(x) + g(x)$ and scalar multiplication $(\alpha f)(x) = \alpha f(x)$.
4. $\mathbb{F}^{m\times n}$, the set of $m \times n$ matrices with entries in $\mathbb{F}$, with entry-wise addition and scalar multiplication.

Example 3.4 (Solution Set of a Homogeneous System)**.** Let $A \in \mathbb{R}^{m\times n}$. The solution set $S = \{x \in \mathbb{R}^n : Ax = 0\}$ is a vector space over $\mathbb{R}$ (a subspace of $\mathbb{R}^n$): if $Ax = 0$ and $Ay = 0$, then $A(x+y) = 0$ and $A(\alpha x) = 0$. In machine learning, the null space of a feature matrix determines the set of weight vectors that produce identical predictions.

3.2 Subspaces

Definition 3.5 (Subspace)**.** A subset $U \subseteq V$ is a *subspace* of V if U is itself a vector space under the operations inherited from V. Equivalently (the *subspace test*), U is a subspace if and only if:

1. $0 \in U$,
2. $u + v \in U$ for all $u, v \in U$ (closure under addition),
3. $\lambda u \in U$ for all $\lambda \in \mathbb{F}$, $u \in U$ (closure under scalar multiplication).

Proposition 3.6 (Intersection of Subspaces)**.** *If U_1, U_2 are subspaces of V, then $U_1 \cap U_2$ is a subspace of V. More generally, the intersection of any collection of subspaces is a subspace.*

Proof. We verify the three conditions of Definition 3.5. (1) Since $0 \in U_1$ and $0 \in U_2$, we have $0 \in U_1 \cap U_2$. (2) If $u, v \in U_1 \cap U_2$, then $u, v \in U_1$, so $u + v \in U_1$ by closure in U_1. Similarly $u + v \in U_2$. Hence $u + v \in U_1 \cap U_2$. (3) If $u \in U_1 \cap U_2$ and $\lambda \in \mathbb{F}$, then $\lambda u \in U_1$ and $\lambda u \in U_2$, so $\lambda u \in U_1 \cap U_2$. The argument for an arbitrary collection is identical, checking each subspace in the family. □

Union of Subspaces

The *union* of two subspaces is generally *not* a subspace. For instance, in $\mathbb{R}^2$ let $U_1 = \{(x, 0) : x \in \mathbb{R}\}$ and $U_2 = \{(0, y) : y \in \mathbb{R}\}$. Then $(1, 0) \in U_1$ and $(0, 1) \in U_2$, but $(1, 1) = (1, 0) + (0, 1) \notin U_1 \cup U_2$. The correct notion that combines two subspaces into a larger subspace is the *sum*, defined next.

3.3 Sums and Direct Sums

Definition 3.7 (Sum of Subspaces)**.** If $U_1, \ldots, U_m$ are subspaces of V, their *sum* is

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_i \in U_i \text{ for each } i\}.$$

Proposition 3.8. *The sum $U_1 + \cdots + U_m$ is the smallest subspace of V containing every U_i.*

Proof. The sum is a subspace: it contains 0 (take each $u_i = 0$); if $u = u_1 + \cdots + u_m$ and $v = v_1 + \cdots + v_m$ with $u_i, v_i \in U_i$, then $u + v = (u_1 + v_1) + \cdots + (u_m + v_m)$ with $u_i + v_i \in U_i$; and $\lambda u = \lambda u_1 + \cdots + \lambda u_m$ with $\lambda u_i \in U_i$. Each U_i is contained in the sum (set all other summands to 0). Any subspace containing every U_i must contain all sums $u_1 + \cdots + u_m$, so the sum is the smallest such subspace. □

Definition 3.9 (Direct Sum)**.** The sum $U_1 + \cdots + U_m$ is called a *direct sum*, written $U_1 \oplus \cdots \oplus U_m$, if each element of the sum can be expressed in *exactly one way* as $u_1 + \cdots + u_m$ with $u_i \in U_i$.

Proposition 3.10 (Characterization of Direct Sums for Two Subspaces)**.** *$V = U_1 \oplus U_2$ if and only if $V = U_1 + U_2$ and $U_1 \cap U_2 = \{0\}$.*

Proof. ($\Rightarrow$) Suppose $V = U_1 \oplus U_2$ and let $v \in U_1 \cap U_2$. Then $0 = v + (-v)$ with $v \in U_1$ and $-v \in U_2$. But also $0 = 0 + 0$ with $0 \in U_1$ and $0 \in U_2$. By the uniqueness requirement in the direct sum, $v = 0$.

($\Leftarrow$) Suppose $V = U_1 + U_2$ and $U_1 \cap U_2 = \{0\}$. If $u_1 + u_2 = u_1' + u_2'$ with $u_i, u_i' \in U_i$, then $u_1 - u_1' = u_2' - u_2$. The left side belongs to U_1 and the right to U_2, so both belong to $U_1 \cap U_2 = \{0\}$. Hence $u_1 = u_1'$ and $u_2 = u_2'$, giving uniqueness. □

Example 3.11 (Direct Sum in $\mathbb{R}^3$)**.** Let $U = \{(x, y, 0) : x, y \in \mathbb{R}\}$ and $W = \{(0, 0, z) : z \in \mathbb{R}\}$. Every $(x, y, z) \in \mathbb{R}^3$ has the unique decomposition $(x, y, 0) + (0, 0, z)$ with the first summand in U and the second in W. Since $U \cap W = \{0\}$, Proposition 3.10 confirms $\mathbb{R}^3 = U \oplus W$.

Theorem 3.12 (Dimension of a Sum)**.** *If U_1 and U_2 are finite-dimensional subspaces of V, then*

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2).$$

Proof. Let $\{w_1, \ldots, w_k\}$ be a basis of $U_1 \cap U_2$. Extend this to a basis $\{w_1, \ldots, w_k, u_1, \ldots, u_p\}$ of U_1 (by Proposition 4.11 below) and to a basis $\{w_1, \ldots, w_k, v_1, \ldots, v_q\}$ of U_2.

We claim the list $\mathcal{L} = \{w_1, \ldots, w_k, u_1, \ldots, u_p, v_1, \ldots, v_q\}$ is a basis of $U_1 + U_2$.

Spanning. Any $x \in U_1 + U_2$ is $x = a + b$ with $a \in U_1$ and $b \in U_2$. Then a is a combination of the w's and u's, and b is a combination of the w's and v's, so x lies in $\operatorname{span}(\mathcal{L})$.

Independence. Suppose $\sum_i \alpha_i w_i + \sum_j \beta_j u_j + \sum_\ell \gamma_\ell v_\ell = 0$. Then $\sum_\ell \gamma_\ell v_\ell = -\sum_i \alpha_i w_i - \sum_j \beta_j u_j$. The left side lies in U_2 and the right side lies in U_1, so both sides belong to $U_1 \cap U_2$.

Hence $\sum_\ell \gamma_\ell v_\ell = \sum_i \delta_i w_i$ for some scalars δ_i. Since $\{w_1, \dots, w_k, v_1, \dots, v_q\}$ is a basis of U_2 (hence linearly independent), all $\gamma_\ell = 0$ and all $\delta_i = 0$. The remaining equation $\sum_i \alpha_i w_i + \sum_j \beta_j u_j = 0$, together with the independence of the basis of U_1, gives all $\alpha_i = 0$ and $\beta_j = 0$.

Therefore $\dim(U_1+U_2) = k+p+q = (k+p)+(k+q)-k = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$. □

Feature Subspaces in Representation Learning

In multi-task learning, each task may learn features in a subspace U_i of a shared representation space. The dimension formula $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$ quantifies how much the tasks share: a large intersection means the tasks exploit similar features, justifying a shared hidden layer with dimension close to $\dim U_1$ rather than $\dim U_1 + \dim U_2$.

With the vocabulary of subspaces and sums in place, we now turn to the building blocks from which every finite-dimensional subspace is constructed: spanning sets and linearly independent lists.

4 Bases and Dimension

4.1 Span

Definition 4.1 (Span)**.** The *span* (or *linear span*) of a list of vectors $v_1, \dots, v_m \in V$ is

$$\operatorname{span}(v_1, \dots, v_m) = \{\lambda_1 v_1 + \cdots + \lambda_m v_m : \lambda_1, \dots, \lambda_m \in \mathbb{F}\}.$$

By convention, $\operatorname{span}(\,) = \{0\}$ (the span of the empty list is $\{0\}$). A vector space V is *finite-dimensional* if there exists a finite list of vectors that spans V.

Example 4.2 (Span in $\mathbb{R}^3$)**.** The vectors $e_1 = (1,0,0)$ and $e_2 = (0,1,0)$ satisfy $\operatorname{span}(e_1, e_2) = \{(a,b,0) : a, b \in \mathbb{R}\}$, which is the xy-plane in $\mathbb{R}^3$. Adding $e_3 = (0,0,1)$ gives $\operatorname{span}(e_1, e_2, e_3) = \mathbb{R}^3$.

4.2 Linear Independence

Definition 4.3 (Linear Independence)**.** Vectors $v_1, \dots, v_m \in V$ are *linearly independent* if the only choice of scalars satisfying $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ is $\lambda_1 = \cdots = \lambda_m = 0$. They are *linearly dependent* otherwise, i.e., some non-trivial combination equals 0.

Example 4.4 (Testing Independence)**.** In $\mathbb{R}^3$, the vectors $(1,0,0)$, $(1,1,0)$, $(1,1,1)$ are linearly independent: if $\alpha(1,0,0) + \beta(1,1,0) + \gamma(1,1,1) = (0,0,0)$, then the system $\alpha + \beta + \gamma = 0$, $\beta + \gamma = 0$, $\gamma = 0$ gives $\gamma = 0$, $\beta = 0$, $\alpha = 0$.

On the other hand, $(1,0,0)$, $(0,1,0)$, $(1,1,0)$ are linearly dependent because $(1,0,0) + (0,1,0) - (1,1,0) = (0,0,0)$.

Lemma 4.5 (Linear Dependence Lemma)**.** *If $v_1, \dots, v_m$ are linearly dependent and $v_1 \neq 0$, then there exists an index $j \in \{2, \dots, m\}$ such that:*

1. $v_j \in \operatorname{span}(v_1, \dots, v_{j-1})$, *and*
2. $\operatorname{span}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) = \operatorname{span}(v_1, \dots, v_m)$.

In words, one can remove v_j without shrinking the span.

Proof. Since the list is linearly dependent, there exist scalars $\lambda_1, \dots, \lambda_m$, not all zero, with $\sum_{i=1}^m \lambda_i v_i = 0$. Let j be the largest index such that $\lambda_j \neq 0$. We must have $j \geq 2$: if $j = 1$, then $\lambda_1 v_1 = 0$ with $\lambda_1 \neq 0$, giving $v_1 = 0$, contradicting $v_1 \neq 0$.

Solving for v_j:

$$v_j = -\lambda_j^{-1} \sum_{i=1}^{j-1} \lambda_i v_i \in \operatorname{span}(v_1, \dots, v_{j-1}).$$

For (2), any vector in $\operatorname{span}(v_1, \ldots, v_m)$ can replace its v_j component using the expression above, so removing v_j does not change the span. □

4.3 Basis and Dimension

Definition 4.6 (Basis)**.** A *basis* of V is a list of vectors in V that is both linearly independent and spans V.

Example 4.7 (Standard Basis of $\mathbb{F}^n$)**.** The *standard basis* of $\mathbb{F}^n$ is $e_1 = (1, 0, \ldots, 0)$, $e_2 = (0, 1, \ldots, 0)$, $\ldots$, $e_n = (0, 0, \ldots, 1)$. Independence is immediate, and any $(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n$.

Example 4.8 (Basis of Polynomial Space)**.** A basis of $\mathbb{F}[x]_{\leq n}$ is $\{1, x, x^2, \ldots, x^n\}$. Thus $\dim \mathbb{F}[x]_{\leq n} = n + 1$.

Theorem 4.9 (All Bases Have the Same Length)**.** *If V is finite-dimensional, then any two bases of V have the same number of elements.*

Proof. Let $\mathcal{U} = \{u_1, \ldots, u_m\}$ and $\mathcal{W} = \{w_1, \ldots, w_n\}$ be two bases of V. We show $m \leq n$.

Since $\mathcal{W}$ spans V, the list $w_1, \ldots, w_n, u_1$ is linearly dependent (it has $n+1$ vectors in a space spanned by n vectors). By the linear dependence lemma (Lemma 4.5), since $w_1 \neq 0$, we can remove some w_j without changing the span. We thus obtain a spanning list of length n that includes u_1.

Repeat: insert u_2 into this list (making it linearly dependent, as it has $n + 1$ elements spanning an n-dimensional space) and remove one of the remaining w's. After m steps, we have a spanning list consisting of all $u_1, \ldots, u_m$ together with $n - m$ of the original w's. This requires $n - m \geq 0$, so $m \leq n$.

Exchanging the roles of $\mathcal{U}$ and $\mathcal{W}$ gives $n \leq m$. Therefore $m = n$. □

Dimension Is Well-Defined

Theorem 4.9 ensures that the following definition is unambiguous: the dimension of a finite-dimensional vector space is the common length of all its bases. This single number encodes the "size" of the space in a coordinate-free way.

Definition 4.10 (Dimension)**.** The *dimension* of a finite-dimensional vector space V, denoted $\dim V$, is the number of elements in any (hence every) basis of V.

Proposition 4.11 (Basis Extension)**.** *Every linearly independent list in a finite-dimensional vector space V can be extended to a basis of V.*

Proof. Let $v_1, \ldots, v_m$ be linearly independent in V with $\dim V = n$. If $m = n$, the list already spans V: a linearly independent list of length equal to the dimension must be a basis (any vector not in the span would extend it to an independent list of length $n + 1$, contradicting Theorem 4.9).

If $m < n$, the list does not span V, so there exists $v_{m+1} \notin \operatorname{span}(v_1, \ldots, v_m)$. Then $v_1, \ldots, v_{m+1}$ is linearly independent (any dependence relation involving v_{m+1} would place v_{m+1} in $\operatorname{span}(v_1, \ldots, v_m)$). Repeat until the list has length n. □

Proposition 4.12 (Dimension of a Subspace)**.** *If U is a subspace of a finite-dimensional space V, then $\dim U \leq \dim V$. Equality holds if and only if $U = V$.*

Proof. Any linearly independent list in U is also linearly independent in V, so its length is at most $\dim V$. Hence $\dim U \leq \dim V$.

If $\dim U = \dim V = n$, let $\{u_1, \ldots, u_n\}$ be a basis of U. This is a linearly independent list of length n in V, so by Theorem 4.9 it is a basis of V. Thus every $v \in V$ lies in $\operatorname{span}(u_1, \ldots, u_n) \subseteq U$, giving $V \subseteq U$ and hence $U = V$.

Conversely, $U = V$ trivially implies $\dim U = \dim V$. □

Embedding Dimension and Feature Rank

In neural network design, the hidden layer dimension d in an encoder $\phi\colon \mathbb{R}^n \to \mathbb{R}^d$ controls the representational capacity. The image $\phi(\mathbb{R}^n)$ (under a linear encoder) is a subspace of $\mathbb{R}^d$ with $\dim(\phi(\mathbb{R}^n)) = \operatorname{rank}(\phi) \leq \min(n, d)$. Choosing d too small forces $\operatorname{rank}(\phi) < n$, losing information. The dimension inequality in Proposition 4.12 makes this precise: the encoder image, as a subspace, cannot exceed the ambient dimension d.

Having established that finite-dimensional vector spaces are characterized up to isomorphism by a single integer (the dimension), we now study the maps that respect vector space structure.

5 Linear Maps

5.1 Definition and Basic Properties

Definition 5.1 (Linear Map)**.** A function $T\colon V \to W$ between vector spaces over $\mathbb{F}$ is a *linear map* if for all $u, v \in V$ and $\lambda \in \mathbb{F}$:

1. $T(u + v) = Tu + Tv$ (additivity),
2. $T(\lambda v) = \lambda Tv$ (homogeneity).

Equivalently, $T(\alpha u + \beta v) = \alpha Tu + \beta Tv$ for all $\alpha, \beta \in \mathbb{F}$ and $u, v \in V$. The set of all linear maps from V to W is denoted $\mathcal{L}(V, W)$.

Example 5.2 (Linear Maps)**.**

1. The *zero map* $0\colon V \to W$ defined by $0(v) = 0$ for all v.
2. The *identity map* $I\colon V \to V$ defined by $Iv = v$ for all v.
3. *Differentiation* $D\colon \mathbb{F}[x]_{\leq n} \to \mathbb{F}[x]_{\leq n-1}$ defined by $Dp = p'$ is linear.
4. For any matrix $A \in \mathbb{F}^{m \times n}$, the map $T_A\colon \mathbb{F}^n \to \mathbb{F}^m$ defined by $T_A(x) = Ax$ is linear.

5.2 Null Space and Range

Definition 5.3 (Null Space and Range)**.** For a linear map $T\colon V \to W$:

- The *null space* (or *kernel*) of T is $\operatorname{null} T = \{v \in V : Tv = 0\}$.
- The *range* (or *image*) of T is $\operatorname{range} T = \{Tv : v \in V\}$.

Proposition 5.4. *For any linear map $T\colon V \to W$, the null space $\operatorname{null} T$ is a subspace of V and the range $\operatorname{range} T$ is a subspace of W.*

Proof. **Null space.** $T(0) = 0$, so $0 \in \operatorname{null} T$. If $u, v \in \operatorname{null} T$, then $T(u + v) = Tu + Tv = 0 + 0 = 0$, so $u + v \in \operatorname{null} T$. If $\lambda \in \mathbb{F}$, then $T(\lambda u) = \lambda Tu = \lambda \cdot 0 = 0$, so $\lambda u \in \operatorname{null} T$.

Range. $T(0) = 0$, so $0 \in \operatorname{range} T$. If $w_1, w_2 \in \operatorname{range} T$, write $w_1 = Tv_1$ and $w_2 = Tv_2$. Then $w_1 + w_2 = T(v_1 + v_2) \in \operatorname{range} T$. If $\lambda \in \mathbb{F}$, then $\lambda w_1 = T(\lambda v_1) \in \operatorname{range} T$. □

Proposition 5.5 (Injectivity Criterion)**.** *A linear map $T\colon V \to W$ is injective if and only if $\operatorname{null} T = \{0\}$.*

Proof. ($\Rightarrow$) If T is injective and $Tv = 0 = T(0)$, then $v = 0$. Hence $\operatorname{null} T = \{0\}$.

($\Leftarrow$) Suppose $\operatorname{null} T = \{0\}$ and $Tu = Tv$. Then $T(u - v) = Tu - Tv = 0$, so $u - v \in \operatorname{null} T = \{0\}$, giving $u = v$. □

5.3 The Rank–Nullity Theorem

The Rank–Nullity Theorem

Theorem 5.6 (Rank–Nullity Theorem)**.** *If V is finite-dimensional and $T\colon V \to W$ is a linear map, then*

$$\dim V = \dim(\operatorname{null} T) + \dim(\operatorname{range} T).$$

Equivalently, writing $\operatorname{nullity}(T) = \dim(\operatorname{null} T)$ *and* $\operatorname{rank}(T) = \dim(\operatorname{range} T)$*:*

$$\dim V = \operatorname{nullity}(T) + \operatorname{rank}(T).$$

Proof. Let $\{u_1, \dots, u_m\}$ be a basis of $\operatorname{null} T$. By Proposition 4.11, extend it to a basis $\{u_1, \dots, u_m, v_1, \dots, v_k\}$ of V, so that $\dim V = m + k$. We claim that $\{Tv_1, \dots, Tv_k\}$ is a basis of $\operatorname{range} T$.

Spanning. Let $w \in \operatorname{range} T$. Then $w = T(x)$ for some $x \in V$. Write $x = \alpha_1 u_1 + \cdots + \alpha_m u_m + \mu_1 v_1 + \cdots + \mu_k v_k$. Applying T and using $Tu_i = 0$:

$$w = T(x) = \mu_1 Tv_1 + \cdots + \mu_k Tv_k.$$

Hence $\{Tv_1, \dots, Tv_k\}$ spans $\operatorname{range} T$.

Independence. Suppose $\mu_1 Tv_1 + \cdots + \mu_k Tv_k = 0$. By linearity, $T(\mu_1 v_1 + \cdots + \mu_k v_k) = 0$, so $\mu_1 v_1 + \cdots + \mu_k v_k \in \operatorname{null} T$. Since $\{u_1, \dots, u_m\}$ is a basis of $\operatorname{null} T$, there exist scalars α_i with

$$\mu_1 v_1 + \cdots + \mu_k v_k = \alpha_1 u_1 + \cdots + \alpha_m u_m.$$

Rearranging: $\alpha_1 u_1 + \cdots + \alpha_m u_m - \mu_1 v_1 - \cdots - \mu_k v_k = 0$. Since $\{u_1, \dots, u_m, v_1, \dots, v_k\}$ is a basis of V (hence linearly independent), all coefficients vanish: $\alpha_i = 0$ for each i and $\mu_j = 0$ for each j.

Therefore $\dim(\operatorname{range} T) = k$ and $\dim V = m + k = \dim(\operatorname{null} T) + \dim(\operatorname{range} T)$. □

Corollary 5.7 (Equivalence for Square Dimensions)**.** *Let $T\colon V \to W$ be linear with $\dim V = \dim W = n$. The following are equivalent:*

(i) T is injective.
(ii) T is surjective.
(iii) T is invertible (bijective).

Proof. By the rank–nullity theorem, $\operatorname{nullity}(T) + \operatorname{rank}(T) = n$.

(i)⇒(ii): Injectivity means $\operatorname{nullity}(T) = 0$, so $\operatorname{rank}(T) = n = \dim W$, i.e., $\operatorname{range} T = W$ (surjectivity).

(ii)⇒(i): Surjectivity means $\operatorname{rank}(T) = n$, so $\operatorname{nullity}(T) = 0$ (injectivity).

(iii)⇔(i)∧(ii): Invertibility is, by definition, injectivity plus surjectivity, and we have shown these are equivalent. □

Example 5.8 (Rank–Nullity for Differentiation)**.** Let $D\colon \mathbb{R}[x]_{\le 3} \to \mathbb{R}[x]_{\le 2}$ be the differentiation map $Dp = p'$. Here $\dim(\mathbb{R}[x]_{\le 3}) = 4$ and $\operatorname{null} D = \operatorname{span}(1)$ (the constant polynomials), so $\operatorname{nullity}(D) = 1$. By rank–nullity, $\operatorname{rank}(D) = 4 - 1 = 3 = \dim(\mathbb{R}[x]_{\le 2})$, confirming that D is surjective (every polynomial of degree ≤ 2 is the derivative of some cubic).

Example 5.9 (Projection Map)**.** Define $P\colon \mathbb{R}^3 \to \mathbb{R}^3$ by $P(x_1, x_2, x_3) = (x_1, x_2, 0)$. Then $\operatorname{null} P = \{(0, 0, z) : z \in \mathbb{R}\}$ with $\operatorname{nullity}(P) = 1$, and $\operatorname{range} P = \{(x, y, 0) : x, y \in \mathbb{R}\}$ with $\operatorname{rank}(P) = 2$. Indeed $1 + 2 = 3 = \dim \mathbb{R}^3$.

Rank Deficiency and Bottleneck Layers

In a deep neural network, each linear layer $W_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$ defines a linear map $\mathbb{R}^{d_{\ell-1}} \to \mathbb{R}^{d_\ell}$. By rank–nullity, $\operatorname{rank}(W_\ell) + \operatorname{nullity}(W_\ell) = d_{\ell-1}$. If $d_\ell < d_{\ell-1}$ (a bottleneck), then $\operatorname{nullity}(W_\ell) \geq d_{\ell-1} - d_\ell > 0$: information is inevitably lost. This is precisely the mechanism exploited in autoencoders, where the bottleneck forces the network to learn a low-dimensional representation.

5.4 Matrix Representation

Fix ordered bases $\mathcal{B} = \{v_1, \ldots, v_n\}$ of V and $\mathcal{C} = \{w_1, \ldots, w_m\}$ of W. A linear map $T\colon V \to W$ is uniquely determined by its action on the basis of V. Write

$$Tv_j = \sum_{i=1}^{m} A_{ij}\, w_i, \qquad j = 1, \ldots, n.$$

The $m \times n$ matrix $A = (A_{ij})$ is called the *matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$*, denoted $\mathcal{M}(T, \mathcal{B}, \mathcal{C})$ or simply $[T]_{\mathcal{B}}^{\mathcal{C}}$.

Proposition 5.10. *If $S\colon U \to V$ has matrix B (with respect to appropriate bases) and $T\colon V \to W$ has matrix A, then $T \circ S$ has matrix AB.*

Proof. Let $\{e_1, \ldots, e_p\}$, $\{v_1, \ldots, v_n\}$, $\{w_1, \ldots, w_m\}$ be the ordered bases of U, V, W respectively. Then $Se_k = \sum_j B_{jk} v_j$ and $Tv_j = \sum_i A_{ij} w_i$. Hence

$$(T \circ S)(e_k) = T\Big(\sum_j B_{jk} v_j\Big) = \sum_j B_{jk} T v_j = \sum_j B_{jk} \sum_i A_{ij} w_i = \sum_i \Big(\sum_j A_{ij} B_{jk}\Big) w_i = \sum_i (AB)_{ik}\, w_i.$$

So the matrix of $T \circ S$ is AB. □

Example 5.11 (Matrix of Differentiation)**.** Let $D\colon \mathbb{R}[x]_{\leq 2} \to \mathbb{R}[x]_{\leq 1}$ be differentiation. With basis $\mathcal{B} = \{1, x, x^2\}$ for the domain and $\mathcal{C} = \{1, x\}$ for the codomain: $D(1) = 0 = 0 \cdot 1 + 0 \cdot x$, $D(x) = 1 = 1 \cdot 1 + 0 \cdot x$, $D(x^2) = 2x = 0 \cdot 1 + 2 \cdot x$. Thus

$$[D]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

5.5 Change of Basis and Similarity

Definition 5.12 (Change-of-Basis Matrix)**.** Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\mathcal{B}' = \{v_1', \ldots, v_n'\}$ be two ordered bases of V. The *change-of-basis matrix* from $\mathcal{B}$ to $\mathcal{B}'$ is the $n \times n$ matrix P whose j-th column is the coordinate vector of v_j with respect to $\mathcal{B}'$:

$$v_j = \sum_{i=1}^{n} P_{ij}\, v_i', \qquad j = 1, \ldots, n.$$

Equivalently, if $[x]_{\mathcal{B}}$ denotes the coordinate vector of x relative to $\mathcal{B}$, then $[x]_{\mathcal{B}} = P\,[x]_{\mathcal{B}'}$ for all $x \in V$... wait, let us be precise. We have $P\,[x]_{\mathcal{B}'} = [x]_{\mathcal{B}}$, i.e., P converts $\mathcal{B}'$-coordinates to $\mathcal{B}$-coordinates.

Remark 5.13. The matrix P is invertible since it represents the identity map $I\colon V \to V$ with respect to the bases $\mathcal{B}'$ (domain) and $\mathcal{B}$ (codomain). Its inverse P^{-1} converts $\mathcal{B}$-coordinates to $\mathcal{B}'$-coordinates.

Proposition 5.14 (Similarity of Matrix Representations)**.** *Let $T\colon V \to V$ be a linear operator. If T has matrix A with respect to basis $\mathcal{B}$ and matrix A' with respect to basis $\mathcal{B}'$, then*

$$A' = P^{-1}AP,$$

where P is the change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$ (so that $P[x]_{\mathcal{B}'} = [x]_{\mathcal{B}}$).

Proof. For any $x \in V$, coordinate representations satisfy:

$$A'[x]_{\mathcal{B}'} = [Tx]_{\mathcal{B}'} = P^{-1}[Tx]_{\mathcal{B}} = P^{-1}A\,[x]_{\mathcal{B}} = P^{-1}A\,P\,[x]_{\mathcal{B}'}.$$

Since this holds for all $x \in V$ and the coordinate map relative to $\mathcal{B}'$ is an isomorphism, $A' = P^{-1}AP$. □

Example 5.15 (Change of Basis in $\mathbb{R}^2$)**.** Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map with matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ relative to the standard basis $\mathcal{B} = \{e_1, e_2\}$. Consider the new basis $\mathcal{B}' = \{(1,0),(1,1)\}$. The change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$ has columns $[v_1']_{\mathcal{B}} = (1,0)^\top$ and $[v_2']_{\mathcal{B}} = (1,1)^\top$:

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$A' = P^{-1}AP = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

The new basis diagonalizes T. This illustrates the general strategy: find a basis of eigenvectors to simplify the matrix representation.

5.6 Isomorphism

Proposition 5.16 (Isomorphism Criterion)**.** *Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.*

Proof. ($\Rightarrow$) Let $T\colon V \to W$ be an isomorphism and let $\{v_1, \dots, v_n\}$ be a basis of V. We show $\{Tv_1, \dots, Tv_n\}$ is a basis of W.

Independence: If $\sum_i \lambda_i T v_i = 0$, then $T(\sum_i \lambda_i v_i) = 0$. Since T is injective, $\sum_i \lambda_i v_i = 0$, and since $\{v_i\}$ is independent, all $\lambda_i = 0$.

Spanning: For any $w \in W$, surjectivity of T gives $w = Tv$ for some $v = \sum_i \lambda_i v_i$. Then $w = \sum_i \lambda_i T v_i \in \operatorname{span}(Tv_1, \dots, Tv_n)$.

Hence $\dim W = n = \dim V$.

($\Leftarrow$) Suppose $\dim V = \dim W = n$. Pick bases $\{v_1, \dots, v_n\}$ of V and $\{w_1, \dots, w_n\}$ of W. Define $T\colon V \to W$ by $Tv_i = w_i$ for each i and extend linearly. This map is well-defined (every $v \in V$ has a unique basis expansion) and linear by construction. It is bijective because it maps a basis to a basis: the same argument as above shows $\{Tv_1, \dots, Tv_n\} = \{w_1, \dots, w_n\}$ is a basis of W. □

Bellman Equation as a Linear System

For a fixed policy π in a finite MDP with n states, the Bellman equation is $V^\pi = r^\pi + \gamma P^\pi V^\pi$, where $V^\pi, r^\pi \in \mathbb{R}^n$ and $P^\pi \in \mathbb{R}^{n\times n}$ is the transition matrix. Rearranging gives $(I - \gamma P^\pi)V^\pi = r^\pi$. The operator $I - \gamma P^\pi$ is a linear map $\mathbb{R}^n \to \mathbb{R}^n$. Since $\gamma < 1$ and all eigenvalues of P^π have modulus at most 1, the eigenvalues of $I - \gamma P^\pi$ are $1 - \gamma\lambda_i \neq 0$, so the map is invertible by Corollary 5.7. Hence the Bellman equation has a unique solution.

6 Exercises

Exercise 6.1 (Subspace Verification). $(*)$
Let $V = \mathbb{R}^3$. Determine which of the following subsets are subspaces. Prove your answer in each case.

(a) $S_1 = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$.
(b) $S_2 = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 1\}$.
(c) $S_3 = \{(x, y, z) \in \mathbb{R}^3 : xz = 0\}$.

Exercise 6.2 (Dimension of a Sum). $(*)$
Let $U = \operatorname{span}\big((1, 1, 0, 0), (1, 0, 1, 0)\big)$ and $W = \operatorname{span}\big((1, 0, 1, 0), (0, 0, 1, 1)\big)$ in $\mathbb{R}^4$. Compute $\dim(U \cap W)$, $\dim(U + W)$, and determine whether $U + W$ is a direct sum.

Exercise 6.3 (Rank–Nullity Application). $(**)$
Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be the linear map defined by

$$T(x_1, x_2, x_3, x_4) = (x_1 + x_2,\ x_2 + x_3,\ x_3 + x_4).$$

(a) Find a basis for $\operatorname{null} T$ and compute $\operatorname{nullity}(T)$.
(b) Find a basis for $\operatorname{range} T$ and compute $\operatorname{rank}(T)$.
(c) Verify the rank–nullity theorem for this map.

Exercise 6.4 (Change of Basis and Diagonalization). $(**)$
Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ have matrix $A = \left(\begin{smallmatrix} 5 & 3 \\ -6 & -4 \end{smallmatrix}\right)$ with respect to the standard basis.

(a) Find the eigenvalues of A.
(b) For each eigenvalue, find an eigenvector and verify it forms a basis of $\mathbb{R}^2$.
(c) Compute the change-of-basis matrix P and verify $P^{-1}AP$ is diagonal.

Hint: The characteristic polynomial is $\lambda^2 - \lambda - 2$.

Exercise 6.5 (RL Application: Feature Matrix Rank). $(**)$
In linear value function approximation, the value of state s_i is approximated as $\hat{V}(s_i) = \phi(s_i)^\top w$ where $\phi(s_i) \in \mathbb{R}^d$ is a feature vector. For n states, define the feature matrix $\Phi \in \mathbb{R}^{n \times d}$ with row i equal to $\phi(s_i)^\top$.

(a) Show that $\hat{V} = \Phi w$ defines a linear map $T_\Phi\colon \mathbb{R}^d \to \mathbb{R}^n$ and that $\operatorname{range} T_\Phi$ is the set of representable value functions.
(b) Prove that if $d < n$ and $\operatorname{rank}(\Phi) = d$, then T_Φ is injective but not surjective: there exist value functions that cannot be represented.
(c) What happens if $\operatorname{rank}(\Phi) < d$? Describe the null space of T_Φ and explain its implications for learning.

Exercise 6.6 (Direct Sum Decomposition). $(***)$
Let V be a finite-dimensional vector space and U a subspace of V. Prove that there exists a subspace W of V such that $V = U \oplus W$.

Hint: Extend a basis of U to a basis of V and let W be the span of the added vectors.

7 Summary

- A *vector space* over $\mathbb{F}$ is a set with addition and scalar multiplication satisfying seven axioms (commutativity, associativity, identities, inverses, distributivity).
- A *subspace* is a nonempty subset closed under addition and scalar multiplication; equivalently, it contains 0 and is closed under linear combinations. The intersection of subspaces is always a subspace, but the union generally is not.

- The *sum* $U_1 + U_2$ is the smallest subspace containing both U_1 and U_2; it is a *direct sum* $U_1 \oplus U_2$ precisely when $U_1 \cap U_2 = \{0\}$.
- A *basis* is a linearly independent spanning list. All bases of a finite-dimensional space have the same length, which defines the *dimension*.
- The *linear dependence lemma* is the engine behind most dimension arguments: in a dependent list with nonzero first vector, some later vector can be removed without changing the span.
- The *rank–nullity theorem* states $\dim V = \operatorname{nullity}(T) + \operatorname{rank}(T)$ for any linear map $T\colon V \to W$ with V finite-dimensional. This is the fundamental constraint relating injectivity, surjectivity, and dimension.
- Every linear map between finite-dimensional spaces has a unique *matrix representation* once bases are chosen. Changing the basis yields a *similar* matrix $A' = P^{-1}AP$.
- Two finite-dimensional vector spaces are *isomorphic* if and only if they have the same dimension.

8 Further Reading

- **Lesson 6** (Eigenvalues and Eigenvectors) builds directly on the matrix representation and similarity theory developed here. The goal is to find bases in which the matrix of a linear operator takes a particularly simple form (diagonal or upper triangular).
- **Lesson 7** (Inner Product Spaces) introduces additional geometric structure – lengths, angles, and orthogonality – on vector spaces. The key results (Gram–Schmidt, orthogonal projections) require the basis and dimension theory from this lesson.
- **Lesson 8** (Spectral Theorem and SVD) combines eigenvalue theory with inner product structure to obtain the spectral theorem for self-adjoint operators and the singular value decomposition.
- For a deeper treatment of infinite-dimensional vector spaces and their applications to ML, see Phase 01 on functional analysis (metric spaces, normed spaces, Hilbert spaces), which generalizes the finite-dimensional theory developed here.
- Axler [2015] is the primary reference for the "determinant-free" approach to linear algebra followed in this lesson. Hoffman and Kunze [1971] provides an alternative, more algebraic perspective. Lax [2007] connects the finite-dimensional theory to operator theory and applications.

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